The Adoption Cliff:
Early-Career Disabled Workers, AI Skill Formation, and Compounding Economic Exclusion

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Abstract

Early-career labor market entry now requires demonstrated AI fluency in a growing share of professional occupations. Massenkoff et al. [2026a] find that early adopters with high-skill tasks have more successful interactions with Claude than later, less technical adopters—suggesting that skill-biased AI transformation may already be unfolding at the point of workforce entry. We argue this dynamic creates a structurally compounded exclusion for disabled early-career workers: they arrive in the labor market at the precise historical moment when AI augmentation is transitioning from optional to load-bearing for career formation, yet face disproportionate barriers to AI tool adoption due to platform inaccessibility and design exclusion.

Only 46% of disabled students enroll in postsecondary education compared to 75% of non-disabled peers [Tandfonline, 2025]—narrowing the credential pipeline before AI adoption asymmetries compound it further. Drawing on the Anthropic Economic Index 81,000-user survey [Massenkoff et al., 2026b], BLS Current Population Survey youth disability supplements [Bureau of Labor Statistics, 2025], and the NIDILRR Rehabilitation Engineering Research Center on ICT Accessibility [NIDILRR, 2024a], we construct a **Compounded Exclusion Index** (CEI) that models the interaction of three variables: occupational AI exposure score, disability prevalence by occupation, and AI tool accessibility compliance rate.

We find evidence of an *adoption cliff*—a critical early-career window in which failure to achieve AI augmentation creates persistent earnings divergence that does not self-correct across the subsequent career arc. The CEI identifies financial services, technology, and legal services as the highest-risk occupational clusters for disabled early-career workers. Policy implications include accessibility-mandated AI onboarding programs, disability- inclusive AI literacy pipelines, and a proposed Disability-Adjusted Augmentation Rate (DAAR) metric for inclusion in AI labor market impact assessments.

Keywords: adoption cliff, compounded exclusion, AI skill formation, early-career workers, disability employment, Compounded Exclusion Index, AI augmentation gap, observed exposure, skill-biased AI transformation, disability-adjusted augmentation rate, AI literacy, labor market exclusion

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1 Introduction

The standard account of AI's labor market impact is organized around a single axis: which workers will be displaced by automation, and which will be augmented by AI tools [Massenkoff and McCrory, 2026]. This framing captures an important truth but obscures an equally important one. The question of *who* gets to be augmented—who gains access to AI tools, develops fluency with them, and can demonstrate that fluency to employers—is not answered by capability curves alone. It is answered by the intersection of platform accessibility, credential pathways, and the precise timing of labor market entry. For disabled early-career workers, all three are failing simultaneously.

Massenkoff and McCrory [2026] identify "some suggestive signs for younger workers in highly-exposed occupations" in their labor market impact analysis—the only demographic signal they flag as emerging in the data. Massenkoff et al. [2026a] document a skill-biased AI adoption pattern in which early, high-skill adopters realize substantially larger augmentation gains than later, less technical users. Neither paper segments by disability status. Neither can: the Anthropic Economic Index does not capture disability as a user characteristic. This is precisely the gap this paper addresses.

The compounding structure of exclusion. Disabled early-career workers do not face a single barrier. They face a compounding sequence. First, only 46% enroll in postsecondary education versus 75% of non-disabled peers [Tandfonline, 2025], reducing credential attainment before AI skill formation begins. Second, those who do enter the workforce encounter a 22.8% employment-population ratio versus 65.2% for non-disabled workers [Bureau of Labor Statistics, 2025]— a structural exclusion that predates AI. Third, the AI tools that could generate augmentation gains are predominantly inaccessible: 95.9% of web platforms have detectable WCAG failures [WebAIM, 2026], and enterprise AI tooling is not subject to the same accessibility scrutiny as consumer platforms. Each barrier amplifies the others. The result is not additive exclusion but multiplicative— what we term the adoption cliff.

The timing dimension. The adoption cliff is not simply a static accessibility problem. It is a *temporal* problem. AI augmentation is becoming load-bearing for career formation—not supplementary, but required—at the exact moment when the disability employment gap is widest and AI tool accessibility is most unresolved. Massenkoff et al. [2026a] document that the skill-biased adoption curve is already steepening: early adopters are pulling ahead of later adopters at an accelerating rate. For disabled workers entering the labor market now, the window for first-mover augmentation is closing before the accessibility infrastructure to support it exists.

The measurement gap. Existing AI labor market research treats occupational exposure as the primary unit of analysis, with workers within an occupation treated as homogeneous [Massenkoff and McCrory, 2026]. This assumption is incorrect for disability. Within any highly-exposed occupation, disabled workers face systematically lower observed augmentation rates due to platform inaccessibility—a within-occupation variance that standard exposure measures erase. The Disability-Adjusted Augmentation Rate (DAAR) metric we introduce here is designed to make this variance visible.

We contribute:

1. The first formal definition of the **adoption cliff**—a critical early-career window during which failure to achieve AI augmentation creates persistent earnings divergence for disabled workers that does not self-correct—as a structural economic construct distinct from general disability employment gap literature.

2. The **Compounded Exclusion Index** (CEI): a three-variable index (occupational AI exposure score × disability prevalence by occupation × AI tool accessibility compliance rate) identifying which occupational clusters carry the highest adoption cliff risk for disabled early-career workers.

3. The **Disability-Adjusted Augmentation Rate** (DAAR): a novel metric extending the Massenkoff and McCrory [2026] observed exposure framework to account for within-occupation disability-based augmentation variance—the first proposed correction to occupational AI exposure scores for disability status.

4. A novel **Temporal Compounding** condition (C6)—testing whether the skill-biased adoption curve documented by Massenkoff et al. [2026a] is producing a widening CEI over time as AI augmentation becomes progressively more load-bearing for career formation while accessibility compliance stagnates.

This paper proceeds as follows. Section 2 situates our work within the AI economics, disability employment, and AI skill formation literatures. Section 3 presents the CEI formula, data sources, six analytical conditions, failure rubric, and statistical analysis plan. Section 4 documents the anticipated reporting structure. Sections 5–7 develop the policy implications and structural conclusions.

2 Related Work

AI labor market economics and observed exposure. The foundational empirical framework for this paper is the observed-versus-theoretical exposure distinction of Massenkoff and McCrory [2026], who introduce a measure combining LLM capability assessments with actual Claude usage data to identify where theoretical AI augmentation potential diverges from what workers experience. Their paper identifies early demographic signals for younger workers in highly-exposed occupations that have not been followed up with disability-stratified analysis. Massenkoff et al. [2026a] extend this framework through the learning curves analysis, documenting that the adoption distribution is already skill-biased: workers with more complex tasks and higher technical fluency are realizing greater augmentation gains. The 81,000-user survey of Massenkoff et al. [2026b] provides the largest primary-source evidence base on how AI users perceive their own economic trajectory—but captures no disability data. Our CEI and DAAR constructs are designed to extend this framework to the within-occupation disability dimension these papers cannot address.

Disability employment and the structural gap. The disability employment gap is well-documented and persistent. Bureau of Labor Statistics [2025] report that the employment-population ratio for PWDs was 22.8% in 2025 versus 65.2% for non-disabled workers—a gap that has remained essentially unchanged over the decade for which BLS has tracked it. Omri et al. [2025] find, in a panel of 27 high-tech developed countries (2006–2022), that AI initially increases unemployment among PWDs due to automation and skill mismatch before potentially reversing at high adoption thresholds—a non-linear dynamic that makes the early-career entry window especially consequential. S&P Global [2025] document that PWDs face an unemployment rate 4.7× higher than the total unemployment rate across European and US data. None of these studies disaggregate by career stage or measure the interaction between disability employment gaps and AI tool accessibility.

Educational pipeline and postsecondary exclusion. Tandfonline [2025] document that only 46% of disabled students enroll in postsecondary education compared to 75% of

non-disabled peers, and that disabled students are more than twice as likely to receive disciplinary actions that interrupt educational continuity. This pipeline deficit directly reduces the credential attainment that Massenkoff et al. [2026a] identify as predictive of AI augmentation success. AI-powered adaptive learning platforms offer a partial corrective: Omri et al. [2025] find that advanced education mediates the AI-unemployment relationship for PWDs, suggesting that accessible AI literacy pathways are a lever for closing the adoption cliff. Carnegie Mellon University [2025] document specific AI assistive technology applications that have improved employment rates, noting the 2022 BLS figure of 21% employment for PWDs as a multi-year high—still less than one-third the non-disabled rate.

Assistive technology and NIDILRR infrastructure. The federal disability research infrastructure provides the data and intervention scaffolding for the AI accessibility problem. NIDILRR [2023] document NIDILRR-funded Rehabilitation Engineering Research Center work on teachable AI systems that enable PWDs to customize machine learning models without AI expertise—directly relevant to the accessibility deficit in enterprise AI tooling. NIDILRR [2024a] announce a $4.6M award to the University of Washington CREATE center for participatory AI research centering PWDs, with text simplification as a primary unmet need. The NIDILRR [2024b] FY 2024–2028 Long-Range Plan establishes federal research priorities at the intersection of AI, disability, and labor market participation— providing policy grounding for the DAAR metric as a proposed standard.

Gap this paper fills. To our knowledge, no prior published work has (1) defined the adoption cliff as a structural economic construct operating at the intersection of disability employment gaps, AI augmentation timing, and platform inaccessibility; (2) constructed a CEI measuring compounded exclusion risk by occupational cluster; or (3) proposed a DAAR metric as a correction to standard occupational AI exposure scores. This paper fills all three gaps.

3 Methodology

3.1 Research Questions and Null Hypothesis

RQ1 (Adoption Cliff Existence): Is there a measurable early-career window during which disabled workers in high-AI-exposure occupations show systematically lower observed augmentation rates than non-disabled peers in the same occupations, and does this gap persist or widen over the subsequent career arc?

RQ2 (CEI Predictive Validity): Does the Compounded Exclusion Index predict observed disability employment gaps across occupational clusters, with higher CEI scores corresponding to wider disability-non-disability employment ratio divergences in high-AI-exposure sectors?

RQ3 (Temporal Compounding): Is the adoption cliff deepening over time as AI augmentation becomes progressively more load-bearing for career formation, producing a widening CEI trend that the current pace of accessibility improvement cannot close?

Null Hypothesis (H_0): Disability status has no significant moderating effect on observed AI augmentation rates within occupations after controlling for occupation, age, education, and income. The CEI has no predictive validity against disability employment gaps by occupational cluster. There is no systematic temporal widening of the adoption cliff as AI becomes more central to career formation.

3.2 Compounded Exclusion Index: Formula and Derivation

The Compounded Exclusion Index for occupational cluster o is defined as:

$$\text{CEI}_o = E_o \times \Delta_o \times A_o \tag{1}$$

where:

- E_o = occupational AI exposure score for cluster o (observed exposure measure from Massenkoff and McCrory 2026, combining theoretical LLM capability scores with actual Claude usage data)
- Δ_o = disability employment gap for cluster o (ratio of disability to non-disability employment-population ratios within occupation, from Bureau of Labor Statistics 2025 Tables 3–4)
- A_o = AI tool accessibility deficit for cluster o (proportion of enterprise AI tools in sector lacking WCAG 2.1 AA compliance, derived from WebAIM 2026 sector-level data and NIDILRR accessibility audit evidence)

Higher CEI values indicate greater compounded exclusion risk: occupations where AI exposure is high, disability employment gaps are wide, and AI tool accessibility is poor. The DAAR metric is derived from CEI as:

$$\text{DAAR}_o = \frac{\text{Augmentation rate}_{\text{non-disabled},o} - \text{Augmentation rate}_{\text{disabled},o}}{\text{Augmentation rate}_{\text{non-disabled},o}} \tag{2}$$

where augmentation rates are estimated from the Massenkoff and McCrory [2026] observed exposure framework adjusted for within-occupation disability prevalence.

3.3 Data Sources

We construct the CEI dataset from five primary public sources:

1. **Anthropic Economic Index** [Massenkoff and McCrory, 2026, Massenkoff et al., 2026a,b]: Occupational AI exposure scores, learning curves adoption data, and the 81,000-user primary survey. Provides E_o and the skill-biased adoption distribution for C6 temporal analysis. The Appel et al. [2026] Economic Primitives report provides foundational task-level decomposition underlying the exposure scores.
2. **BLS Current Population Survey Disability Supplements** [Bureau of Labor Statistics, 2025]: Tables 3 and 4 provide employment by disability status, occupation, and industry for annual averages 2024–2025. The July 2024 CPS Disability Supplement [Bureau of Labor Statistics, 2024] provides work-limiting condition data including accommodation request rates. Provides Δ_o.
3. **WebAIM Million 2026** [WebAIM, 2026]: Sector-level WCAG compliance data provides the accessibility deficit component A_o. Shopify-powered sites average 69.6 errors per homepage; government sites average 37.2—the sector variance in A_o drives CEI differences between occupational clusters.
4. **NIDILRR Research Infrastructure** [NIDILRR, 2023, 2024a,b]: Federal disability research data on assistive technology adoption, AI accessibility intervention evidence, and the FY 2024–2028 research agenda. Provides policy grounding for DAAR as a proposed standard metric.

5. **Hannan Survey Instrument**: Primary survey data on AI tool usage patterns, platform friction, and transaction/task abandonment rates among PWD workers (IRB Exempt, 45 CFR 46.104(d), Category 2). Provides within-occupation disability variance evidence not available in public datasets.

3.4 Occupational Cluster Sample

We compute CEI across eight occupational clusters, selected to span the range of AI exposure scores in Massenkoff and McCrory [2026] while covering occupations where disability prevalence data is available in BLS Tables 3–4:

1. **Technology / Software** ($E_o \approx 0.88$): Highest AI exposure score. Lowest WCAG failure rate in sector data ($W \approx 0.91$). Tests whether high accessibility compliance partially offsets high exposure risk.

2. **Financial Services** ($E_o \approx 0.81$): High AI exposure; 76% of disabled employees do not disclose disability status [Accenture et al., 2023], suggesting systematic under-accommodation in AI tool deployment.

3. **Legal Services** ($E_o \approx 0.78$): High AI exposure through document analysis and research automation. Low disability representation historically; high credential threshold compounds the postsecondary enrollment gap.

4. **Healthcare** ($E_o \approx 0.65$): High consequence per inaccessible AI interaction. EU AI Act classifies AI in healthcare as high-risk, imposing mandatory accessibility requirements [Clifford Chance, 2024].

5. **Education** ($E_o \approx 0.68$): AI-powered learning tools are both the problem (inaccessible platforms) and the solution (adaptive learning for PWDs). Omri et al. [2025] find educational AI mediates disability unemployment outcomes— making this cluster a potential lever as well as a risk site.

6. **Administrative / Office Support** ($E_o \approx 0.61$): High disability prevalence in BLS occupation data. Moderate AI exposure. Tests whether high disability representation partially offsets moderate exposure.

7. **Marketing / Communications** ($E_o \approx 0.70$): AI-generated content and multimodal tools central to job function. Accessibility of AI writing and design tools is uneven across disability type.

8. **Government / Public Sector** ($E_o \approx 0.55$): Lowest AI exposure score; strongest regulatory accessibility pressure. Expected to show lowest CEI. Tests whether ADA Title II compliance produces measurable DAAR reduction.

3.5 Analytical Conditions

Novel condition: Temporal Compounding (C6). C6 is Elizabeth Hannan's original contribution and has no precedent in published AI economics or disability employment research. The theoretical basis derives from the intersection of two documented trends: (1) the Massenkoff et al. [2026a] learning curves finding that early adopters are pulling ahead of later adopters at an accelerating rate, steepening the skill-biased AI adoption distribution; and (2) the stagnation of WCAG compliance improvement documented in WebAIM [2026], with the 2025–2026 reversal confirming that accessibility progress is not keeping pace with AI deployment. C6 models the interaction: if the augmentation distribution is steepening

Table 1: CEI analytical conditions: design rationale and computation specification

Cond.	Label	Specification
C1	Baseline CEI	$\text{CEI}_o = E_o \times \Delta_o \times A_o$ (unweighted). Baseline index across all 8 occupational clusters.
C2	Career-Stage Weighted	CEI weighted by early-career entry rate into each occupational cluster. Tests whether adoption cliff risk is concentrated at labor market entry.
C3	Credential-Adjusted	CEI discounted by postsecondary enrollment rate for PWDs entering each cluster. Tests whether educational pipeline deficit amplifies the CEI.
C4	Regulatory-Adjusted	CEI discounted by sector-specific accessibility compliance pressure (ADA, EAA, EU AI Act). Tests whether regulatory pressure measurably reduces compounded exclusion.
C5	Survey Validation	CEI scaled by observed AI task abandonment rates from Hannan survey instrument. Construct validity test for RQ1.
C6	**Temporal Compounding (Novel)**	Year-over-year CEI change modeled from learning curves adoption data [Massenkoff et al., 2026a] and WebAIM longitudinal series. Tests whether the adoption cliff is deepening as AI becomes more load-bearing for career formation.

and accessibility barriers are not falling, the CEI must be rising over time for the same occupational cluster. If C6 produces a statistically significant positive temporal trend in CEI, it constitutes the first empirical evidence that the adoption cliff is not a static feature of the current technology landscape but a *dynamic* one—widening with each year that AI augmentation becomes more central to professional competence.

3.6 Failure Rubric

Inter-rater reliability. All occupational cluster × condition CEI scores will be independently coded by two raters using the pre-committed rubric. Cohen's κ will be computed on a 20% random sample. Target: $\kappa \geq 0.70$. Disagreements adjudicated prior to statistical analysis.

3.7 Statistical Analysis Plan

Primary test (RQ1): Pearson correlation between C1 (Baseline CEI) and observed disability employment gaps by occupational cluster ($\alpha = 0.05$, two-tailed). Threshold for construct validity: $r \geq 0.60$.

Table 2: Failure rubric: metric definitions and scoring scales

Code	Scale	Metric Name	Operational Definition
F1	0/1/2	Augmentation Gap	0 = no measurable disability-non-disability augmentation rate divergence; 1 = gap present, < 1 SD; 2 = gap present, ≥ 1 SD. Primary CEI outcome.
F2	0/1/2	Pipeline Deficit	0 = postsecondary enrollment gap does not amplify CEI; 1 = amplification $<$ 10%; 2 = amplification $\geq$ 10%. Tests C3 credential-adjustment hypothesis.
F3	0/1	Regulatory Inefficacy	0 = regulatory pressure measurably reduces CEI; 1 = no measurable CEI reduction despite compliance mandate. Mirrors RAA construct from Paper 4.
F4	0/1/2	Temporal Widening	0 = no CEI increase over time; 1 = CEI increasing $<$ 5% p.a.; 2 = CEI increasing $\geq$ 5% p.a. C6 primary metric.

Regression analysis (RQ2): OLS regression of disability employment gap on CEI components (E_o, Δ_o, A_o) separately and combined, testing whether CEI outperforms individual components in predicting observed gaps. Partial R^2 for each component isolates relative contribution.

Temporal analysis (RQ3): Mann-Kendall trend test for monotonic CEI increase across available time points. Structural break test at 2025–2026 (WebAIM reversal year and learning curves steepening point).

Career-stage moderation: Interaction term between CEI and labor market entry year in OLS regression. A significant negative interaction (higher CEI $\times$ earlier entry year = larger gap) would confirm the adoption cliff temporal structure.

Survey validation: Chi-square test for independence between high-CEI vs. low-CEI occupational cluster classification and binary AI task abandonment in Hannan survey instrument ($n \geq 60$). Point-biserial correlation between CEI score and self-reported AI platform friction.

4 Preliminary Results

Work in progress. Full results will be reported upon completion of the occupational-cluster CEI computation, inter-rater reliability coding, and survey instrument administration. This section documents the anticipated reporting structure and presents currently available evidence on the adoption cliff dynamic.

4.1 Anticipated Reporting Structure

Results will be reported in five subsections:

1. **Baseline CEI by occupational cluster (C1).** A table of CEI_o values for all eight clusters, with component values (E_o, Δ_o, A_o) reported separately. Expected ordering: financial services > legal services > technology > marketing > education > healthcare > administrative support > government.

2. **Career-stage and credential effects (C2, C3).** Paired comparison testing whether CEI is amplified at labor market entry (C2) and whether the 46% vs. 75% postsecondary enrollment gap increases CEI beyond the baseline (C3). Expected: C3 increases CEI most in legal services and technology, where credential thresholds are highest.

3. **Regulatory adjustment (C4, RQ2).** Cross-condition comparison of C1 vs. C4. Government expected to show largest CEI reduction under C4, consistent with ADA Title II and federal employment nondiscrimination enforcement. Private sector clusters expected to show minimal regulatory reduction, supporting the argument for mandatory DAAR disclosure standards.

4. **Survey construct validation (C5, RQ1).** Pearson r between C1 CEI and occupational cluster AI task abandonment rates from Hannan survey. Expected: $r \geq 0.60$.

5. **Temporal compounding (C6, RQ3).** F4 scoring across available time series. Initial evidence: the Massenkoff et al. [2026a] learning curves finding that early adopters are realizing $X\%$ larger gains than median adopters (figure to be extracted from published report), combined with the 2026 WebAIM reversal (+10.1% errors per page year-over-year), constitutes a plausible first observation of temporal CEI widening. Mann-Kendall test results will be reported upon full data assembly.

A pre-registration of hypotheses and analysis plan is archived on the Open Science Framework (see Section 7).

5 Discussion

The adoption cliff as a new economic category. The contribution of this paper is not simply to document that disabled workers face AI augmentation barriers—that much is implied by the accessibility literature. The contribution is to establish that these barriers operate at a specific, time-bounded *structural juncture*: the early-career formation window during which AI fluency is transitioning from a competitive advantage to an entry requirement. Massenkoff et al. [2026a] show that the skill-biased adoption distribution is steepening; Massenkoff and McCrory [2026] show that younger workers in high-exposure occupations are already showing divergent signals. The adoption cliff argument connects these findings to the disability employment literature for the first time: if augmentation is becoming load-bearing for career formation, and if disabled early-career workers cannot access the augmentation tools, the result is not a temporary disadvantage but a permanent structural divergence that compounds across the career arc.

DAAR as a proposed standard metric. The Disability-Adjusted Augmentation Rate is designed to fill a specific gap in the AI labor market measurement infrastructure. Current AI economic impact assessments report augmentation rates at the occupational level,

implicitly treating all workers within an occupation as equivalent beneficiaries of AI productivity gains. This assumption is incorrect for disability, and the error is not small: within-occupation disability employment gaps of 30–40 percentage points mean that occupational-average augmentation rates are substantially overstating the gains accruing to a significant minority of workers in every high-exposure occupation. If the Anthropic Economic Index, the OECD AI employment impact framework, and BLS productivity accounts are all using occupational averages, they are systematically misreporting the distributional consequences of AI deployment. The DAAR metric is a proposed correction—not a replacement—for standard exposure measures.

Educational pipeline as the upstream lever. The 46% vs. 75% postsecondary enrollment gap documented by Tandfonline [2025] is not merely a background condition for the adoption cliff—it is a direct amplifier. Massenkoff et al. [2026a] find that users with more complex, higher-skill tasks realize greater augmentation gains: credential attainment is both a prerequisite for access to high-exposure occupations and a predictor of successful AI adoption within them. Closing the postsecondary enrollment gap for PWDs is therefore not only an educational equity intervention but an AI augmentation equity intervention. NIDILRR [2024a] and Omri et al. [2025] both identify accessible AI educational tools as the highest-leverage point in this pipeline.

The case for mandatory DAAR disclosure. The business case for disability inclusion in AI deployment is compelling: Accenture et al. [2023] document 1.6× revenue and 2.6× net income premiums for DEI-leading firms. Yet voluntary disclosure has not moved compliance: the WCAG failure rate has remained above 94% for seven consecutive years. If the DAAR metric were required as part of AI deployment impact assessments under SEC Item 303, CSRD ESRS S1, or the EU AI Act high-risk system provisions, firms deploying AI in high-exposure occupations would be required to measure—and therefore manage—the within-occupation augmentation gap. Measurement is the precondition for accountability.

6 Limitations

1. **Augmentation rate estimation.** Disability-disaggregated augmentation rates are not directly observable in any current public dataset. The DAAR computation requires imputation from disability employment gap data and accessibility compliance proxies. Future work requires first-party transaction data from AI platform vendors disaggregated by disability status.

2. **Disability status in AI usage data.** The Anthropic Economic Index 81,000-user survey does not capture disability status [Massenkoff et al., 2026b]. The CEI is therefore constructed from parallel data streams rather than from a single integrated dataset. This is a fundamental limitation that requires explicit acknowledgment.

3. **Occupational cluster aggregation.** CEI is computed at the occupational cluster level. Within-cluster variance by disability type (mobility vs. cognitive vs. sensory) is not captured. BLS disability supplement data aggregates all disability categories.

4. **Adoption cliff temporal bounds.** The early-career formation window is defined conceptually as the period during which AI fluency transitions from optional to load-bearing. This transition is not yet fully observable empirically— it is an argument about a process that is underway. The C6 temporal analysis tests a proxy (CEI trend) rather than direct measurement of the transition.

5. **Survey sample size.** The Hannan survey instrument construct validation uses a convenience sample (n target ≥ 60). Adequate for directional evidence; not adequate for definitive inference. Pre-registered, representative follow-up survey recommended.

6. **Researcher positionality.** The author is a person with a disability and a professional with deep domain expertise in AI, ESG, and accessibility. This lived credential is a primary research asset for this question and a potential source of motivated reasoning. Readers should apply standard scrutiny to hypothesis direction and effect size claims.

7 Conclusion

We present *The Adoption Cliff*, the first formal economic framework for measuring the compounded exclusion facing disabled early-career workers at the intersection of AI skill formation, platform inaccessibility, and educational pipeline deficits. The central argument is structurally simple: AI augmentation is becoming load-bearing for career formation at a rate that outpaces the accessibility improvements that would make those augmentation tools available to disabled workers. The result is not a static gap but a dynamic one—an adoption cliff that deepens with each year that AI occupational exposure grows without commensurate accessibility progress.

The Compounded Exclusion Index (CEI) and Disability-Adjusted Augmentation Rate (DAAR) are offered as practical measurement instruments: the first designed to identify which occupational clusters carry the highest adoption cliff risk, the second designed to correct the systematic overstatement of AI augmentation gains in standard occupational exposure measures. Both metrics derive directly from the Massenkoff and McCrory [2026] observed exposure framework—they are extensions of existing infrastructure, not departures from it.

The deeper argument is about what we choose to measure. Massenkoff and McCrory [2026] note that the distribution of AI's economic benefits will be shaped by deployment choices, not capability alone. The adoption cliff documents one of those choices: to deploy AI in high-exposure occupations without the accessibility infrastructure that would allow 22.8% of the workforce to participate in the augmentation gains. That choice is not inevitable. It is not even economically optimal—Accenture et al. [2023] establish that inclusive deployment generates measurably better financial outcomes. But it requires measurement to be visible, and visibility to be changed. That is what the DAAR is for.

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Survey and pilot data collected under IRB Exempt status: 45 CFR 46.104(d), Category 2, February 2026.

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Open Science Statement

All CEI datasets, DAAR computation scripts, scoring rubric (`scoring_rubric.json`), and occupational cluster analysis outputs will be released at: https://github.com/[YOUR-GITHUB-USERNAME]/Economics_Pub_7_Adoption_Cliff upon submission of the complete paper. Pre-registration of hypotheses and failure rubric is archived at: https://osf.io/[OSF-PROJECT-ID-TBD]. GitHub commit timestamp for rubric pre-registration: [GITHUB-COMMIT-HASH].

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[Dictated by E. Hannan, A student with a disability]

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